

Problem 1. Asymptotic Relationships

(a)

$f(n)$	$g(n)$	Relationship?
6^n	n^{1200}	$f \in \omega(g)$
$(\log n)^6$	$\sqrt[1200]{n}$	$f \in o(g)$
$\log_6(n^{1200})$	$(\log_{1200} n)^6$	$f \in o(g)$
$\log(n!)$	$n \log n$	$f \in \Theta(g)$
$2^{\lceil \log_2 n \rceil}$	$3^{\lceil \log_3 n \rceil}$	$f \in \Theta(g)$
$2^{\lceil \log_2 n \rceil} - n + 1$	$3^{\lceil \log_3 n \rceil} - n + 1$	None.
n	$n^{\cos \log n}$	None.

(b)

$f(n)$	$g(n)$	Relationship?	Justification
$2^{(\log n)^2}$	$2^{(\lceil \log n \rceil)^2}$	$f \in O(g)$	$\lceil \log n \rceil \geq \log n$
4^n	$(2n)!/(n!)^2$	$f \in \omega(g)$	$\lim_{x \rightarrow \infty} \frac{f(n)}{g(n)} = \infty$

Problem 2. OEIS

(a)

$$x_3 = 5, x_4 = 12, x_5 = 29, x_6 = 70$$

(b)(c) Found formula:

$$a(n) = \frac{(1 + \sqrt{2})^n - (1 - \sqrt{2})^n}{2\sqrt{2}}$$

(d)

Proof by strong induction. Induction hypothesis which says, sequence that begins with $x_0 = 0, x_1 = 1, x_2 = 2$ and continues with $x_n = 5x_{n-2} + 2x_{n-3}$ for $n \geq 3$, nth term can also be calculated via $P(n) = \frac{(1+\sqrt{2})^n - (1-\sqrt{2})^n}{2\sqrt{2}}$ for $x \geq 0$

Base cases: $P(0) = 0$

$$P(1) = \frac{(1 + \sqrt{2})^1 - (1 - \sqrt{2})^1}{2\sqrt{2}} = 1$$

$$P(2) = \frac{(1 + \sqrt{2})^2 - (1 - \sqrt{2})^2}{2\sqrt{2}} = 2$$

Inductive step: For strong induction, assume $P(n-1), P(n-2)$ is true for $n \geq 2$, and we want to show that $P(n+1)$ is also true. By the definition of the given sequence, we know that

$$\begin{aligned}
x_{n+1} &= 5x_{n-1} + 2x_{n-2} \\
&= 5 \left(\frac{(1+\sqrt{2})^{n-1} - (1-\sqrt{2})^{n-1}}{2\sqrt{2}} \right) + 2 \left(\frac{(1+\sqrt{2})^{n-2} - (1-\sqrt{2})^{n-2}}{2\sqrt{2}} \right) \\
&= \frac{1}{2\sqrt{2}} \left[\left(5(1+\sqrt{2})^{n-1} + 2(1+\sqrt{2})^{n-2} \right) - \left(5(1-\sqrt{2})^{n-1} + 2(1-\sqrt{2})^{n-2} \right) \right] \\
&= \frac{1}{2\sqrt{2}} \left[(1+\sqrt{2})^{n-2} [7 + 5\sqrt{2}] - (1-\sqrt{2})^{n-2} [7 - 5\sqrt{2}] \right] \\
&= \frac{1}{2\sqrt{2}} \left[(1+\sqrt{2})^{n-2} [1 + 3\sqrt{2} + 3(\sqrt{2})^2 + (\sqrt{2})^3] - (1-\sqrt{2})^{n-2} [1 - 3\sqrt{2} + 3(\sqrt{2})^2 - (\sqrt{2})^3] \right] \\
&= \frac{1}{2\sqrt{2}} \left[(1+\sqrt{2})^{n-2} [1 + \sqrt{2}]^3 - (1-\sqrt{2})^{n-2} [1 - \sqrt{2}]^3 \right] \\
&= \frac{(1+\sqrt{2})^{n+1} - (1-\sqrt{2})^{n+1}}{2\sqrt{2}}
\end{aligned}$$

Which implies $P(n+1)$, and by the strong induction hypothesis, $P(n)$ holds for all $n \geq 0$. $\square$

Problem 3. Bounded Asymptotics

(a)

Proof. Function f is bounded from above therefore, $\exists B \in \mathbb{R} (f(n) \leq B)$ and since $f : \mathbb{N} \rightarrow \mathbb{R}^+ (B > 0)$. And the function g is bounded away from zero, meaning $\exists A \in \mathbb{R} (A > 0 \text{ and } g(n) \geq A)$. Now let c be a constant real number such that $c = \frac{B}{A}$ and $n_0 = 1$, for all $n > n_0$, then,

$$\begin{aligned}
g(n) &\geq A \quad (\text{multiplying both side by } c) \\
c \cdot g(n) &\geq A \cdot c \\
c \cdot g(n) &\geq B \geq f(n) \\
c \cdot g(n) &\geq f(n)
\end{aligned}$$

Therefore, by the definition of Big O , we can conclude that $f(n) \in O(g(n))$. $\square$

(b) No it is not true

Proof by counterexample. $f(n) = 1$, which is bounded above by value 1, and $g(n) = 1/n$, which is bounded from below by value 1, but here clearly $f(n) \notin O(g(n))$ $\square$

(c) Yea it is true that $f \in O(g)$ because we can always take $f(1)/g(1)$ as value of constant c and multiply with $g(n)$ and since f is strictly decreasing function and $g(n)$ is strictly increasing, condition $f(n) \leq c \cdot g(n)$ will always hold.

No, it is not always true that $f \in o(g)$. For example, $f(n) = 1 + (1/n)$ is strictly decreasing and $g(n) = 2 - (1/n)$ is strictly increasing, but the limit of their ratio will be $\frac{1}{2} \neq 0$.

Proof. $\square$

Problem 4. Karatsuba's Algorithm

(a)

$$T(2) = 3T(1) + 10 * (2) = 3 + 20 = 23$$

$$T(3) = 3T(2) + 10 * (3) = 69 + 30 = 99$$

$$T(4) = 3T(2) + 10 * (4) = 69 + 40 = 109$$

(b)

$$\begin{aligned}
 T(n) &= 3T\left(\frac{n}{2}\right) + 10n \\
 &= 3 \left[3T\left(\frac{n}{2^2}\right) + 10\frac{n}{2} \right] + 10n \\
 &= 3^2 T\left(\frac{n}{2^2}\right) + 3 \cdot 10\frac{n}{2} + 10n \\
 &= 3^2 \left[3T\left(\frac{n}{2^3}\right) + 10\frac{n}{2^2} \right] + 3 \cdot 10\frac{n}{2} + 10n \\
 &= 3^3 T\left(\frac{n}{2^3}\right) + 3^2 \cdot 10\frac{n}{2^2} + 3 \cdot 10\frac{n}{2} + 10n \\
 &\vdots \\
 &= 3^k T\left(\frac{n}{2^k}\right) + 3^{k-1} \cdot \frac{10n}{2^{k-1}} + 3^{k-2} \cdot \frac{10n}{2^{k-2}} + \dots + \frac{10n}{2^0} \\
 &= 3^{\log_2 n} T(1) + \left[\left(\frac{3}{2}\right)^{k-1} + \left(\frac{3}{2}\right)^{k-2} + \dots + \left(\frac{3}{2}\right)^0 \right] 10n \quad \left(\frac{n}{2^k} = 1, k = \log_2 n\right) \\
 &= 3^{\log_2 n} + \left[\frac{\left(\frac{3}{2}\right)^k - 1}{\frac{1}{2}} \right] 10n \\
 &= 3^{\log_2 n} + \left[\frac{\left(\frac{3}{2}\right)^{\log_2 n} - 1}{\frac{1}{2}} \right] 10n \\
 &= 3^{\log_2 n} + \left[\frac{3^{\log_2 n} - n}{\mathcal{N}} \right] 20\mathcal{N} \\
 &= 21 \cdot 3^{\log_2 n} - 20n
 \end{aligned}$$

(c) For Karatsuba's Algorithm $f(n) = 10n$, $a = 3$ and $b = 2$, so for master theorem value of $\log_b^a = \log_2^3$ and $f(n) \in O(n^{\log_2^3})$ because $\log_2^3 > 1$, which implies $T(n) = \Theta(n^{\log_2^3})$ by following the case I of master theorem.

(d) Yes, it is. From part (c), we know Karatsuba's algorithm takes $\Theta(n^{\log_2^3})$, and the value of $\log_2^3 \approx 1.58$, meaning it takes $\Theta(n^{1.58})$ steps, which is an improvement over the straightforward algorithm which takes $\Theta(n^2)$ steps.